

A Power User's Desktop Guide

Covers Ubuntu 9.10
& 10.04, Includes DVD



Ubuntu

Up and Running

O'REILLY®

Robin Nixon

Ubuntu: Up and Running

Ubuntu for everyone! This popular Linux-based operating system is perfect for people with little technical background. It's simple to install and easy to use—with a strong focus on security.

Ubuntu: Up and Running shows you the ins and outs of this system with a complete hands-on tour. You'll learn how Ubuntu works, how to quickly configure and maintain Ubuntu, and how to use this unique operating system for networking, business, and home entertainment.

This book includes a DVD with the complete Ubuntu system and several specialized editions—including the Mythbuntu multimedia release. The DVD contains a 9.10 distribution that is easy to update to 10.04.

- Choose among three desktop environments: GNOME, KDE, or XFCE
- Connect printers, sound cards, cameras, webcams, and iPhones
- Use the Linux filesystem with either the desktop or the command line
- Learn networking functions, such as file and folder sharing and Internet access
- Get an in-depth introduction to the OpenOffice.org business productivity suite
- Learn how to use WINE to run Windows programs
- Implement upgrades quickly and easily
- Find out where to get and how to install other great software for Linux

“Robin’s efforts combined with the reputation and publishing prowess of O’Reilly contribute another brick in the wall toward mainstream Ubuntu success.”

—Jono Bacon

Ubuntu community manager

Robin Nixon has worked with and written about computers since the early 1980s. He developed the first known pop-up windows and has worked for many of Britain’s main IT magazine publishers. Robin is the author of *Learning PHP, MySQL, and JavaScript* (O’Reilly).

You should understand the basics of computing and be familiar with using an operating system such as Windows or OS X.

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Robin Nixon

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Ubuntu: Up and Running

by Robin Nixon

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For Julie.

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Foreword

When I first heard about Ubuntu back in 2004, a few years before becoming the Ubuntu Community Manager, I was captivated by the approach that Mark Shuttleworth, the founder of Ubuntu, was taking with his new venture. Not only did he have a firm commitment to rock-solid technology, forged by a team of open source rock stars with Debian as a foundation, but he underlined his vision with similarly rock-solid commitment to community.

At the time, the industry growing up around open source was still fairly new, Linux was beginning to be commercialized, and community was increasingly seen as an impediment in companies with a traditional software development culture. Corporations and community were beginning to clash, and it was easier for many companies to merely tolerate community rather than embrace it.

Things seemed different with Ubuntu and its primary sponsor, Canonical. Ubuntu took a confident and adventurous approach to their new operating system. They made opinionated decisions about the best tools to satisfy given use cases, integrated exciting new technologies before others were willing to do so, and delivered the whole shebang on a single CD to make distribution and installation easy.

Ubuntu was compelling from a technical and usability standpoint—which in itself was exciting—but as a community management dork, what really excited me was the open and transparent approach to community that Shuttleworth adopted from the project's very launch. Right out of the gate, Ubuntu provided open communication and development resources, an openly governed community council and technical board, public meetings, and even open specification tracking for each new release. This was stunning. Most companies at that time were regressing in the areas of openness and transparency, not progressing. This is when that cheeky young upstart called Ubuntu came screaming into my own consciousness, first as a curiosity, then as an interest, and ultimately as a passion to grow a global, inclusive, and energized community around an ethos of openness and rocking software.

At the time this book is published, I will have been at Canonical for over three and a half years, and the journey we have been on has been remarkable. We have seen our small community grow into global, sprawling user and contributor bases that transcend

the borders of countries, languages, and cultures. As our community has grown, we have been united by a shared set of values and energized by a vision of just what could be possible if we came together to change the status quo. What excited me three and a half years ago about Ubuntu was its willingness to question and challenge the norms of the time, underlined by a code of ethics that put community at the heart of the approach. That sense of adventure has never been so effervescent as it is today, and we now have a diverse global movement united by the same spirit; each set of eyes, ears, and fingers putting their brick in the wall to effect change in the way we approach computers, information, and freedom. This is what community is all about: a shared dream in which we all play a part.

This book is one such contribution to that dream. Robin has taken on quite a challenge and put together a comprehensive guide to the ins and outs of Ubuntu. As Robin was writing this book, we were in the thick of our latest release cycle. Robin worked hard to stay ahead of the curve, writing and explaining the operation of functionality that was sometimes only a few days old. I did this once myself when I wrote a book, and it is a complex, time-consuming, and sometimes frustrating process, but Robin took to it with grace and elegance. Robin's efforts, combined with the reputation and publishing prowess of O'Reilly, contribute another brick in the wall toward mainstream Ubuntu success.

I am hugely proud of the progress that we have made so far in our journey, but we still have plenty of road ahead of us. While our global Ubuntu family has broken down many barriers and achieved many things, the fast-changing world of technology brings forward new challenges and opportunities. Fortunately, the firm foundation of our community and drive to succeed will help us to scale these challenges and celebrate the opportunities. Before we step into the future, though, let us celebrate our achievements of today by exploring the many features and nuances in Ubuntu outlined and explained in Robin's book. I hope to see you all joining us on our continued journey!

—Jono Bacon
Ubuntu community manager

Preface

Ubuntu is far and away the most popular version (or distribution) of the Linux operating system. Dreamed up by a dot-com billionaire and ex-cosmonaut, and based on years of work by thousands of dedicated programmers, Ubuntu is a serious contender in both the desktop and the web server worlds.

It is based on the philosophy that software should be free, both economically and legally, in that you should be able to install it at no cost, and should also be allowed to modify and/or distribute it without paying any fees or royalties, which is achieved by releasing Ubuntu under a special license that protects these rights.

This means you may install Ubuntu on any computer on which it will run, and can upgrade to the latest version as often as you like, without worrying about licenses, product activation, or special keys you have to enter. You can give it to your friends and family too.

Ubuntu is also very easy to install, automatically recognizing and setting up the best drivers for the devices you use. And, with other operating systems taking anything from one hour to half a day to install, Ubuntu really shines with an average installation time of under an hour.

For over a decade, Linux has provided the computing power behind millions of web servers, so it's no surprise with the amount of work that has been put into developing graphical frontends, desktop installations of Linux now exceed 1% of the total (a figure that is rapidly growing), with Ubuntu taking the lion's share of that amount.

Everything you need to know to become a seasoned Ubuntu user is in this book, which also covers upgrading to and using the latest 10.04 Lucid Lynx release. What's more, it also shows you how to easily and quickly upgrade to the even newer versions that are released every six months.

Audience

This book is for people interested in learning about Ubuntu, the most popular Linux distribution. This includes students, home and business users, hobbyists, professionals, and all other types of user. Because this book takes you from the first steps of installing

the operating system, through configuration, maintenance, and security, and on to topics such as networking, it provides a complete resource for all beginning to intermediate users.

Assumptions This Book Makes

This book assumes that you have a basic understanding of your computer and are comfortable using either Windows or Mac OS X, but no prior knowledge of Unix, Linux, or Ubuntu is required.

Organization of This Book

The chapters in this book are carefully ordered so that different aspects of the operating system are fully introduced before moving on to the next step. To get the best results from this book, you are therefore recommended to read the chapters in sequence.

For example, to get you started, the first chapter introduces Ubuntu, explains its background, and explores the different parts that make up the operating system. Then, with introductions out of the way, the second chapter takes you straight into installing Ubuntu. The process is fully explained in simple terms with plenty of screen grabs to guide you.

After that, the next chapter shows how to configure Ubuntu to your computer, including setting up your keyboard, mouse, and monitor, and managing peripherals such as printers, sound cards, and webcams.

With your setup now configured, the next chapters introduce the Linux filesystem and how to use it, both with the desktop and via the command line. You are then shown how to install and remove programs with the easy-to-use built-in tools, as well as how to maintain your system and keep it secure.

The following two chapters then cover networking functions such as file and folder sharing and accessing the Internet.

By this time, you will have a solid understanding of Ubuntu, so the remaining chapters show you what you can do with it, such as using the OpenOffice.org office suite of programs, installing and playing games, connecting peripherals such as scanners and Bluetooth devices, running Windows programs, and how to install other flavors of Ubuntu.

Along the way, each new topic is introduced using plain English, with the absolute minimum of jargon, and is supported by screen grabs showing exactly what you should see on your computer.

Supporting Books

Once you have mastered Ubuntu Linux, you may be interested in learning more about Linux and its other distributions. You can take your skills to the next level using the following books, which range from beginner to advanced level:

- *Linux in a Nutshell, Sixth Edition* (<http://oreilly.com/catalog/9780596154493>) by Ellen Siever, Stephen Figgins, Robert Love, and Arnold Robbins (O'Reilly)
- *Bash Cookbook: Solutions and Examples for bash Users* (<http://oreilly.com/catalog/9780596526788>) by Carl Albing, JP Vossen, and Cameron Newham (O'Reilly)
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Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, file extensions, pathnames, directories, and Unix utilities.

Constant width

Indicates command-line options, variables and other code elements, HTML tags, macros, the contents of files, and the output from commands.

Constant width bold

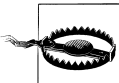
Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values.



This icon signifies a tip, suggestion, or general note.



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Acknowledgments

A huge thank you goes to my editor, Andy Oram, for all his guidance and the tremendous amount of work and all the extra snippets of advice and information he put into making this book such a useful guide. And, as usual, the production staff at O'Reilly also did a magnificent job copyediting the text and designing the book's format and layout. It's a pleasure to work with you all.

With a book introducing something as complex as an operating system, it's essential for it to be as accurate as possible, so I am also very grateful to the team of eagle-eyed technical reviewers: Lesley Harrison, Matthew Helmke, and Jeff Kite, who provided many excellent suggestions for improving the manuscript.

And, of course, this book wouldn't exist without Canonical, GNU, Linux, and the many other organizations that have worked tirelessly for the best part of two decades to integrate a myriad of different parts into the whole known as Ubuntu. Because they number in the thousands, I don't want to single out just a few of the contributors; instead, on behalf of all Ubuntu users, I would simply like to thank each and every person who has participated in creating such an amazing piece of software.

About Ubuntu

Some people say South African entrepreneur Mark Shuttleworth is a very lucky man. Others say he's an astute businessman and talented software engineer. But whichever way you look at it, there's no denying that he has twice helped steer the course of technological development throughout the world.

In 1995, he founded Thawte, the first company outside the United States to produce a fully encrypted, commercially available web server, and a leading provider of server certification, that he sold for a cool half billion dollars just four years later.

Using the proceeds, not only did Shuttleworth achieve his ambition of blasting into space on a Russian Soyuz and spending over a week in the International Space Station, he also funded the development of Ubuntu, now the largest Linux distribution by a long way.

Since its release in 2004, Ubuntu has consistently topped the chart of the most popular Linux distributions, and it currently accounts for over a third of all installations. In 2009, according to *Dimensional Research/KACE*, Ubuntu was the second most likely operating system companies intended to deploy in place of Windows Vista or 7. And if it continues its phenomenal growth, 2010 should see Ubuntu knock OS X out of first place in that poll.

So this book is about Mark Shuttleworth's second major technology hit because, with an estimated 10 million users and growing, a hit it surely is. But to understand what Ubuntu is you need to know a little of its development because it's the result of combining many different unique parts, whose roots can be traced back over 25 years.

Why Ubuntu?

The one question I am most asked is, "Why Ubuntu?" There's no single answer to this. Instead, I offer a selection of reasons depending on the asker's computing requirements. So here, in no particular order, are my reasons why you should choose Ubuntu as your operating system.

Its development is open

The whole philosophy behind the GNU project of which Linux is a part (and Ubuntu is a flavor of Linux) is that “software should be free.” Although that often means the software is free of charge (not always, as you can see in the successful marketing of Red Hat products and Novell’s SUSE products, both based on Linux), it means more importantly that all development is visible and shared.

Downloads and upgrades are free

So whether you are a personal user running Ubuntu at home or one of hundreds of people in a company running Ubuntu, you pay nothing for the software. People around the world, of every economic class, so long as they have access to a computer, can use everything offered as part of Ubuntu. And not only is the original download free but so is every upgrade that follows.

It’s quick to install

If you haven’t installed Ubuntu before but have installed an operating system such as Windows XP, Vista, or 7, you’ll be amazed at how quick and easy Ubuntu is to install.

Upgrading is easy

Whenever a new version is released, Ubuntu will tell you and offer to upgrade automatically, at no cost, as long as you have an Internet connection.

Support is readily available

For home and private users, a wealth of support is available at the Ubuntu website and in forums across the Internet. If you have a problem, you can usually get an answer within hours (if not minutes). This is because Ubuntu is written by volunteers all around the globe who maintain contact with each other via these forums, and they are always happy to help out if they can.

Enterprise support is priced modestly

Although free support is available, businesses will probably find it much more convenient to take out a multi-installation license for a Long Term Support version of Ubuntu (such as 10.04). This brings a whole new level of assistance to the enterprise user, while still keeping the overall running costs of Ubuntu far cheaper than proprietary operating systems such as Windows.

You can modify it

If you have programmers who are familiar with Linux in your organization, they can obtain the Ubuntu source code totally free of charge and can then modify it in any way necessary, as long as they release the new code under the same terms. This can drastically speed up a company’s development cycle by relying on software that’s already been written. This is, in fact, how new features often get added to Linux, and how parts of the operating system are improved.

It’s portable

You can run Ubuntu from a CD-ROM or USB thumb drive without installing it.

It comes with office applications

Ubuntu comes with OpenOffice.org preinstalled so that you can get up and running straight away, writing documents, spreadsheets, and presentations, and your files will be compatible with Microsoft Office.

Of course, there are many other reasons, but these are probably the main ones.

The Parts That Make Up Ubuntu

Ubuntu, named after a South African word meaning “humanity toward others,” is a free operating system (OS) with a strong focus on usability and ease of installation. It is sponsored by the UK company Canonical Ltd., owned by Mark Shuttleworth.

By keeping Ubuntu free and open source (I’ll define these terms in a moment), Canonical says it is able to utilize the talents of a community of developers. Canonical makes its profit from selling technical support and from creating other services related to Ubuntu.

In 2005 Shuttleworth also created the Ubuntu Foundation, kicking it off with a 10 million dollar grant, which he calls insurance for the project should he ever cease his involvement. The foundation’s purpose is to ensure the support and development for all future versions of Ubuntu. In December 2009, Shuttleworth stepped down as CEO of Canonical to “focus on product design, partnerships, and customers.”

Ubuntu (whose logo you can see in [Figure 1-1](#)) strongly relies on other developers, too, as it’s a fork of the Debian project’s code base, itself a popular Linux distribution.



Figure 1-1. The official Ubuntu logo

Debian

Initially developed in 1993, Debian—a contraction of the first name of the developer, Ian Murdock, along with that of Debra Lynn, a former girlfriend—is not backed by any company, yet it still manages to provide the basis of over a dozen other Linux distributions, and is available as both a desktop and a server operating system.

The Debian project is a volunteer organization with over a 1,000 member developers, each working on different aspects of the OS.

GNU

Both Debian and Ubuntu, and most of their constituent parts, are released under the GNU General Public License (GPL) and the Lesser General Public License (LGPL). This is a free software license, meaning that anyone can share the software with other people and any derived works must be available under the same terms.

The term GNU is a recursive acronym (a kind of joke that is popular among software hackers); it stands for GNU's Not Unix. The GNU project was started in 1984 at MIT by Richard Stallman, who had the goal of making a totally free operating system. This was finally achieved in 1992 with the appearance of the Linux kernel, itself released under the GPL.

The Linux Kernel

The Linux kernel is the core operating system used by all distributions of Linux, and is generally pronounced *lin-uks*. It was created in 1991 by Linus Torvalds, a Finnish software engineer, who also had a vision of creating a free operating system.

Because Linux filled a critical gap in the GNU project by providing a working kernel, the total package is often called GNU/Linux. But most people just refer to it as Linux, which I'll do for the rest of this book.

So far, thousands of programmers have worked on the Linux kernel, with even Microsoft contributing 20,000 lines of code. In fact, so much work has gone into it that in 2006 a European Union study put the redevelopment cost of the kernel at over a billion U.S. dollars.

Tux

You may often see a penguin character called Tux used as the Linux logo (see [Figure 1-2](#)). It was created by Larry Ewing using GIMP (the GNU Image Manipulation Program) as an entry to a Linux competition. Surprisingly, it didn't win, but it has since become the official mascot.

Linux Distributions

Linux comes in a variety of different versions called *distributions*, or *distros* for short. Generally each distribution consists of the Linux kernel along with libraries and utilities from the GNU project, as well as a variety of applications such as word processors, spreadsheets, media players, and so on.

Unbelievably, there are estimated to be over 600 distributions of Linux, of which 300 are actively maintained. Of course, the vast majority of them are quite specialized and don't have many users. However, a significant number of distributions are relatively popular, although none of them comes close to the usage of Ubuntu.

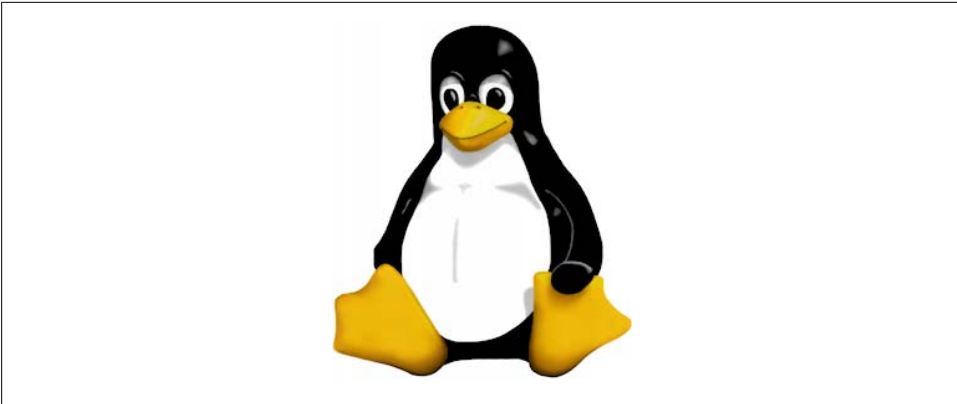


Figure 1-2. Tux, the friendly Linux mascot

Table 1-1 lists some of the more common distributions, roughly in order of popularity according to the <http://distrowatch.com> website. Of course there are too many to list them all, so please don't be disappointed if you don't see all of your favorites.

Table 1-1. Some of the more popular Linux distributions

Name	Type of distribution
Ubuntu	A Linux distribution using the GNOME desktop environment
Fedora	A community distribution sponsored by Red Hat
Mint	A distribution based on and compatible with Ubuntu
openSUSE	Originally derived from Slackware and sponsored by Novell
Debian	A volunteer-driven, noncommercial distribution
Mandriva	A Red Hat derivative popular in France and Brazil
PCLinuxOS	A derivative of Mandriva
Puppy	A very small distribution that fits in 64 MB of memory
Sabayon	A Gentoo-based distribution
Arch	An independent distribution targeted at competent Linux users
CentOS	A 100% compatible rebuild of Red Hat Enterprise Linux
MEPIS	A Debian-based desktop Linux distribution
Slackware	One of the first Linux distributions
Kubuntu	A version of Ubuntu using the KDE desktop environment
Gentoo	A distribution with FreeBSD similarities, targeted at power users
Ubuntu Studio	A version of Ubuntu modified for intensive audiovisual work
Xubuntu	A version of Ubuntu using the Xfce Desktop environment
Red Hat	A derivative of Fedora maintained by Red Hat
Edubuntu	A derivative of Ubuntu designed for use in classrooms and schools

One reason for all these different distributions is that they are usually tailored for specific purposes, such as types of users, ranging from beginner to power user, or for types of computers, ranging from low- to high-end. There are also distributions that focus on education, music, graphics, office, or other tasks.

Some distributions come with the absolute minimum of additional software and a tightly reduced kernel, making them suitable for use in embedded environments such as cell phones or routers, whereas others include every conceivable feature.

Ubuntu Versions

Ubuntu treads the middle line. As can be seen from browsing through [Table 1-1](#), there are only a few officially sanctioned versions of Ubuntu, of which the following are covered in this book:

Ubuntu

The main Ubuntu Linux distribution

Kubuntu

A version that uses the KDE desktop instead of GNOME

Xubuntu

A version that uses the Xfce desktop instead of GNOME

Mythbuntu

A version tailored for use as a personal video recorder (PVR)

Netbook Edition

A version redesigned for use on Netbook computers (previously known as Netbook Remix)

X Window System

Graphics in Ubuntu are displayed thanks to the X Window System, which is separate from Linux and runs on many other operating systems as well. It's free and open source, like Ubuntu.

The X Window System was invented in the 1980s at MIT, explaining the terse name. Computer programmers sometimes avoid fanciful labels, to the point where two of the most popular programming languages are named C and R. The X Window System was derived from another one called W, which ran on an operating system from Stanford called V.

Although X has powered the graphics on nearly all Unix systems (and Unix-like systems, including Linux) for over two decades, its development faltered for a while during debates over licensing and leadership. Luckily, a group of dedicated developers reinvigorated the window with the nonprofit X.org, which was responsible for the code on a new, community-centered footing.

In testimony to the importance of healthy community support for free software, X has undergone a pretty thorough rewrite from the ground up and now supports quite a range of sophisticated graphics hardware. It lies at the basis of all the desktops described in the next section.

Linux Desktop Environments

If you use Windows or a Mac, you will generally access your computer from a graphical desktop environment such as Windows XP, Vista, or 7, or Mac OS X. On Ubuntu Linux, there are three main desktop environments to choose from: GNOME, KDE, and Xfce.

GNOME

GNOME is the default desktop used by Ubuntu. It was designed to provide a working environment with a heavy emphasis on simplicity, usability, and “making things just work.”

The GNOME project arose because the more popular desktop environment at the time, KDE, was built on a development framework called Qt. At the time, this framework wasn’t licensed under the GPL, and presented potential conflicts of interest.

Therefore, GNOME was built entirely using GPL- and LGPL-licensed software and is an example of many different projects being brought together. Based on the GTK+ toolkit, its look and feel is not too dissimilar from Windows or Mac OS X, with moveable windows that can be resized, a launcher menu, taskbar, and notification area (see [Figure 1-3](#)). Other distributions that use GNOME include Debian and Fedora.

KDE

The KDE desktop is based on the Qt framework, which now *is* licensed under the LGPL. It is the desktop environment used by Kubuntu and was first released in 1998 as a modern Unix desktop that gave all applications a consistent look and feel. In 2006 Mark Shuttleworth became the first patron of KDE, the highest level of sponsorship available.

The main difference between Ubuntu and Kubuntu is that, by default, Ubuntu uses GNOME applications, such as the Evolution Personal Information Manager (PIM) and the Synaptic Package Manager, whereas Kubuntu uses the KDE equivalents, such as the Kontact PIM and KPackageKit package manager. There are also numerous programs written for KDE, but you can usually run them from GNOME if you prefer.

However, it is possible to run both KDE and GNOME on the same machine, because Kubuntu and Ubuntu share all the same repositories. You can even install Ubuntu and then add KDE to it, as shown in [Figure 1-4](#). This also has the effect of adding several KDE programs to your GNOME menus, and vice versa.



Figure 1-3. A typical Ubuntu desktop using GNOME

Although your desktop of choice is a personal matter, many users feel that KDE provides a little more power in terms of options and configuration than GNOME, but at the slight expense of organization and user simplicity. Hence KDE is more likely to be installed by more experienced users. Windows users may also find its “K” menu system, located in the bottom-lefthand corner, reminiscent of the Start Menu.

Xfce

The Xfce desktop environment is used by Xubuntu Linux (as well as two other Unix-based operating systems: Solaris and BSD) and is based on the same GTK+ toolkit as GNOME, but it uses the Xfwm window manager.

The Xfce philosophy is “small is simple,” as you can see in Figure 1-5. Due to its ability to run on low specification equipment, it is most commonly implemented on systems with old or resource-limited hardware. Its configuration is mouse-driven, with all advanced options hidden from casual users.

Xfce does, however, include an option to preload the libraries for GNOME and KDE, resulting in the ability to mix applications more quickly than the other major desktops. Although it's extremely fast, some have noted that the Xubuntu desktop is slower than other Xfce implementations.

Xfce is the least used of the three environments, with less than 10% of all Linux desktop installations. That said, it is quite similar to the Windows XP “Classic” desktop and probably deserves more users than it has.

In the same way that KDE is available with Kubuntu, Xfce comes with Xubuntu. But either can be quite easily added to the main Ubuntu distribution, and I'll show you how to do this in [Chapter 2](#).

Linux Packages

Each distribution is usually split up into smaller packages, some of which become optional add-ons that can be installed using a package manager. There are different managers for different flavors of Linux; Ubuntu uses the Synaptic Package Manager (see [Figure 1-6](#)) and, more recently, the Ubuntu Software Center.

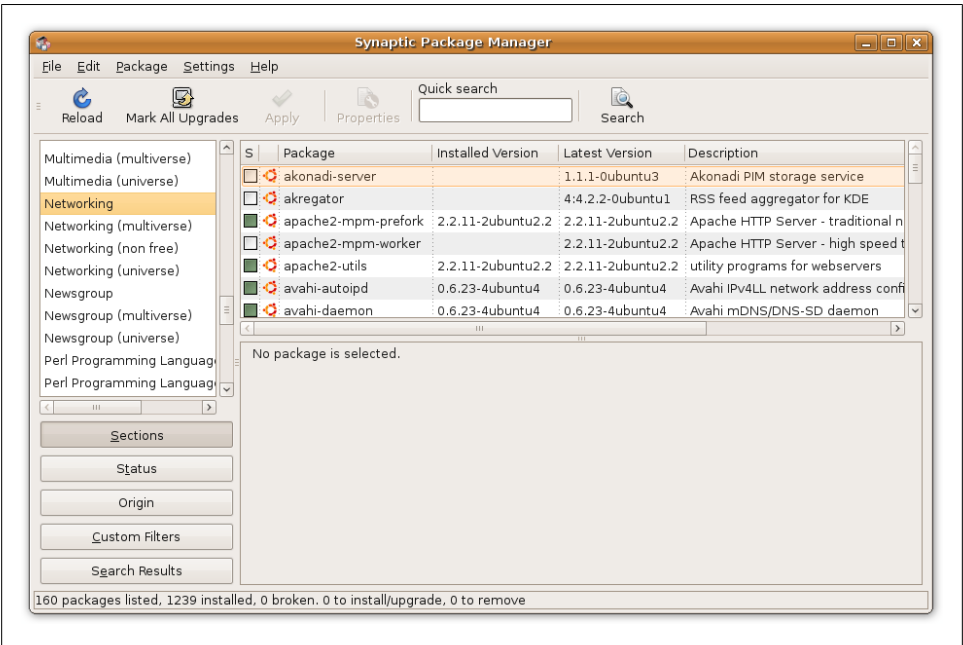


Figure 1-6. The Synaptic Package Manager

If you have heard of people having difficulty installing Linux software, don't be put off, because package managers have changed all that. You can now install almost any program available for Ubuntu with only a few mouse clicks.

The Ubuntu Community

One of the great things about Ubuntu is that it is open source, meaning that not only is it free to use, it's also free to modify, as long as any modifications you make are released under the same license. This means that the thousands of independent developers working on it have a vested interest in sharing all the latest information with one another. When one developer has a problem, another steps in to help, and vice versa.

But more than that, there's a vibrant and growing community that includes Ubuntu's millions of users. Of course they all communicate with each other via email, instant messages, and list servers, etc., but one of the main meeting points is the Ubuntu Community at <http://ubuntu.com/community> (see Figure 1-7), where everyone is welcome to join in. Many members also sign the Ubuntu Code of Conduct (<http://ubuntu.com/community/conduct>), in which they agree to be considerate and respectful of all Ubuntu users and to collaborate with them. So you won't simply be told to "refer to the documentation" when you ask a question in a forum.

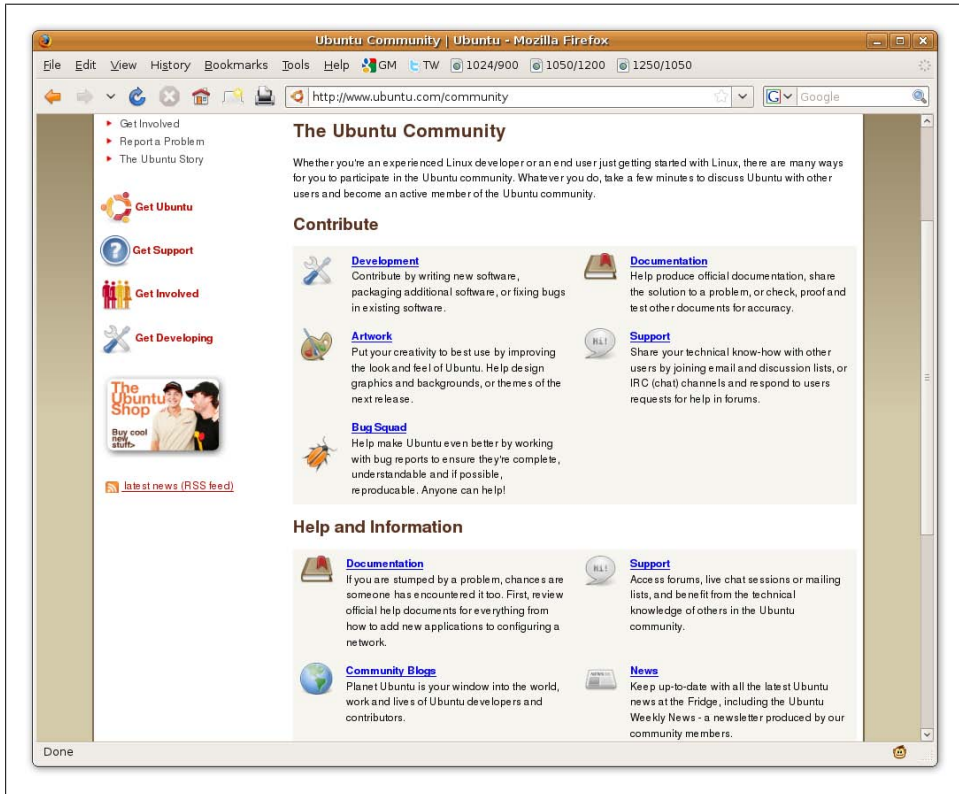


Figure 1-7. The Ubuntu Community website

On the website, you'll find forums covering every aspect of using Ubuntu you can imagine and where, due to its popularity, you can often get an answer to a problem within just a few minutes. There's also full documentation for all the latest releases, the Planet Ubuntu blog, and The Fridge news website.

You can also get more involved in the community and participate by adding to the program code or documentation, or by contributing artwork, user support, or bug reports. Whatever it is, the Ubuntu Community will welcome you.

Ubuntu Versions

There are two releases of Ubuntu each year, generally around April and October. Each release has a major number based on the year and a minor one based on the month. Ubuntu releases are also given code names based on animals, which since the third release have progressed in alphabetical order. A lot of fun precedes each launch, with Ubuntu users everywhere trying to guess whether the next version will be Observant Ostrich or Obvious Otter, and so on. The release history so far is shown in [Table 1-2](#).

Table 1-2. The Ubuntu release history

Version	Code name	Release date	Supported until	
			Desktop	Server
4.10	Warty Warthog	October 2004	April 2006	
5.04	Hoary Hedgehog	April 2005	October 2006	
5.10	Breezy Badger	October 2005	April 2007	
6.06 LTS	Dapper Drake	June 2006	July 2009	June 2011
6.10	Edgy Eft	October 2006	April 2008	
7.04	Feisty Fawn	April 2007	October 2008	
7.10	Gutsy Gibbon	October 2007	April 2009	
8.04 LTS	Hardy Heron	April 2008	April 2011	April 2013
8.10	Intrepid Ibex	October 2008	April 2010	
9.04	Jaunty Jackalope	April 2009	October 2010	
9.10	Karmic Koala	October 2009	April 2011	
10.04 LTS	Lucid Lynx	April 2010	April 2013	April 2015

Things You See

When using Ubuntu, the things you will see on the desktop include applications such as the OpenOffice.org suite of programs (which are compatible with Microsoft Office), the Firefox web browser, or the Terminal (used for entering command-line instructions).

There are also loads of games, graphic manipulation tools, instant messaging utilities, audio-visual applications, and a lot more that I will introduce over the following chapters.

Things You Don't See

What you won't see by default in Ubuntu are important operating system files, program source code, configuration files, and the like. Of course, these are all necessary parts of an operating system and are all there, but they are hidden from casual view, unless you know where to look—in which case, you will be more than a casual user.

This means that on the whole you can rummage around and play with what you find in the Ubuntu desktop, update programs, configure peripherals, and so on, all from easy-to-use graphical frontends. This doesn't mean that you can't make major changes to your computer, because you can, just as you can with Windows or OS X. But it does mean you won't need to roll up your sleeves and get under the desktop's hood too often to configure Ubuntu.

Summary

If any of the information in this chapter has confused you, don't worry. The most important aspect of Ubuntu, and possibly the greatest contribution to its success, is the operating system's usability. It really is as easy as Windows or OS X to install and set up (in fact, it's usually a lot quicker), and you don't actually need to know most of the preceding details.

As the following chapters will show, Ubuntu is a very friendly OS that will not only save you money (because it is free), but will also grow with you due to the tremendous user support and frequent upgrades provided.

I promise that by the time you finish this book, using Ubuntu will be second nature.

Installing and Upgrading Ubuntu

In this chapter, we'll take a look at the companion DVD for this book, which is attached to the inside back cover, and I'll also show readers without access to the DVD how to get the same software by downloading and burning ISO disc images.

Unlike most program discs you may have used, this DVD is unusual in that you don't actually have to install anything if you don't wish to, because it's been set up as a *Live DVD*.

This means that the DVD itself is a fully working installation of Ubuntu 9.10 that will run from the disc without the need to install anything. Therefore you can test Ubuntu to ensure it will work properly on your computer before installing it. It achieves this trick by creating a RAM disc out of system memory, which it then treats as if it were the hard disk.

The disc also contains the installer for Ubuntu 9.10 desktop edition and a set of ISO files for several of the other popular editions of Ubuntu, such as Kubuntu and Xubuntu. This chapter shows you how to use and install these files.



To access the disc, you must have a DVD-compatible optical drive. If you don't, you cannot use the disc and should refer to the section [“Downloading a Distribution” on page 24](#).

The Live DVD

The great thing about the Live DVD system is that you can insert the disc into most PCs and run Ubuntu without installing anything. Once you've verified that all is in order, you can then choose to install Ubuntu, either as the only operating system or as an additional OS (alongside Microsoft Windows, for example). Or you can simply run the Live DVD if you choose.

The downside of running from the Live DVD is that it is much slower than a hard disk installation because DVD drives are not as fast. Also, because the operating system exists only in RAM, any settings you make or documents you save locally will be lost when you close a session (although there are techniques you can use to keep session data on a USB drive, explained at <https://help.ubuntu.com/community/LiveCD/Persistence>). But I recommend you perform a full installation as soon as you're ready. First, however, let's look at simply running Ubuntu from the disc.

To do this, you should ensure that your BIOS settings allow booting from your CD/DVD drive. Systems usually do this by default. If not, the manual that came with your PC will have advice on how to change it. Sometimes you can simply watch the messages that come up on your screen as you boot your system, and they will tell you how to enter BIOS and change your boot settings by pressing a certain key combination.



The DVD will work on an Intel Macintosh computer, but you will need to force it to boot from the DVD by either setting the System Preference for the startup disk, or shutting down and turning off your Mac and then holding down the C key while you turn it on again.

Now insert the disc into your drive and restart your computer. You should be presented with the screen shown in [Figure 2-1](#).

This is where you select the language that should be used for the Live DVD (or during installation if you select that option). Only the main language type is required. For example, as long as English is your native language, go ahead and select it, regardless of which country you are in. Don't worry about local differences such as keyboard layout. In the next section you'll see how you can further tailor the language to your locale.

So either select the language of your choice or, if you wait 30 seconds, English will be chosen by default. Upon selection, the screen shown in [Figure 2-2](#) will appear.

The Main Menu

From here, you have five main choices and a range of options selectable via the function keys. The choices are:

Try Ubuntu without any change to your computer

As it sounds, this choice launches Ubuntu ready for you to use. It then runs from the DVD without modifying your computer's hard disk.

Install Ubuntu

Select this choice when you wish to install Ubuntu on your computer.

Language			
Amharic	Galego	Norsk bokmål	中文(简体)
Arabic	Gujarati	Norsk nynorsk	中文(繁體)
Asturianu	Hebrew	Punjabi (Gurmukhi)	
Беларуская	Hindi	Polski	
Български	Hrvatski	Português do Brasil	
Bengali	Magyar	Português	
Bosanski	Bahasa Indonesia	Română	
Català	Italiano	Русский	
Čeština	日本語	Sámegillii	
Cymraeg	ქართული	Slovenčina	
Dansk	aza	Slovenščina	
Deutsch	Khmer	Shqip	
Dzongkha	한국어	Српски	
Ελληνικά	Kurdî	Svenska	
English	Lietuviškai	Tamil	
Esperanto	Latviski	Thai	
Español	Македонски	Tagalog	
Euskara	Malayalam	Türkçe	
Suomi	Marathi	Українська	
Français	Nepali	Tiếng Việt	
Gaeilge	Nederlands	Wolof	

F1 Help F2 Language F3 Keymap F4 Modes F5 Accessibility F6 Other Options

Figure 2-1. The first screen you see after the Live DVD boots

Check disc for defects

If you are experiencing problems with the DVD and suspect it may be faulty, you can test it using this choice.

Test memory

Using this choice, you can check that all your computer's memory is in working order.

Boot from first hard disk

You will most likely use this choice if you accidentally leave the DVD in your hard drive during a reboot but actually intended to run the operating system installed on your hard drive. With this choice you can quit from the DVD and commence a regular boot from your hard disk.

The Function Key Options

Using these options, you can set up various features before selecting one of the main five menu choices. Most of these options either let you revisit options that were presented to you already or provide workarounds for unusual situations advanced users



Figure 2-2. The Live DVD main menu screen

face, so I'll just point you in the right direction and let you investigate further if necessary.

F1 Help

This brings up a comprehensive help system, the top screen for which is shown in [Figure 2-3](#). From within the help system, you use the function keys shown to select different sections, which you can scroll through with the cursor keys. To exit, press the Esc key.

F2 Language

This option redisplay the language selection menu shown in [Figure 2-1](#).

F3 Keymap

This option is where you can tell Ubuntu about the country in which you live, and therefore which keyboard type you are using, as shown in [Figure 2-4](#).

F4 Modes

With this option you can tell Ubuntu the graphic mode to use, as shown in [Figure 2-5](#). Normally you should leave the default selected unless you experience problems, in which case try the Safe graphics mode. You are unlikely to need either of the remaining two options.

F5 Accessibility

This option provides a number of assistive features such as high contrast, a magnifier, use of a screen reader or Braille terminal, and more. You can see the options available in [Figure 2-6](#).

F6 Other Options

You are extremely unlikely to need this option, which provides a range of different parameters for booting Ubuntu, as shown in [Figure 2-7](#). If you are curious, the help system provides more information on what they do. Purists who do not want to run any proprietary software at all, such as binary device drivers, may wish to select the “Free software only” option.

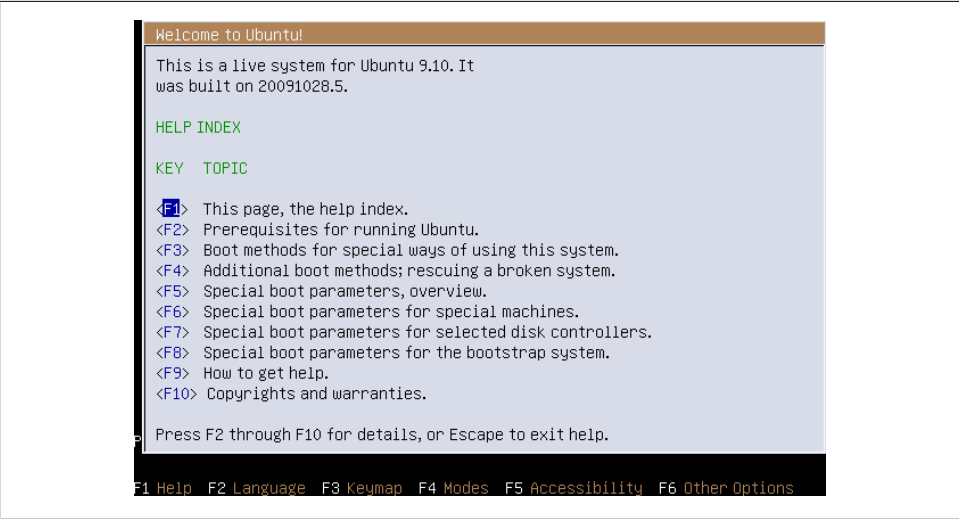


Figure 2-3. The help system

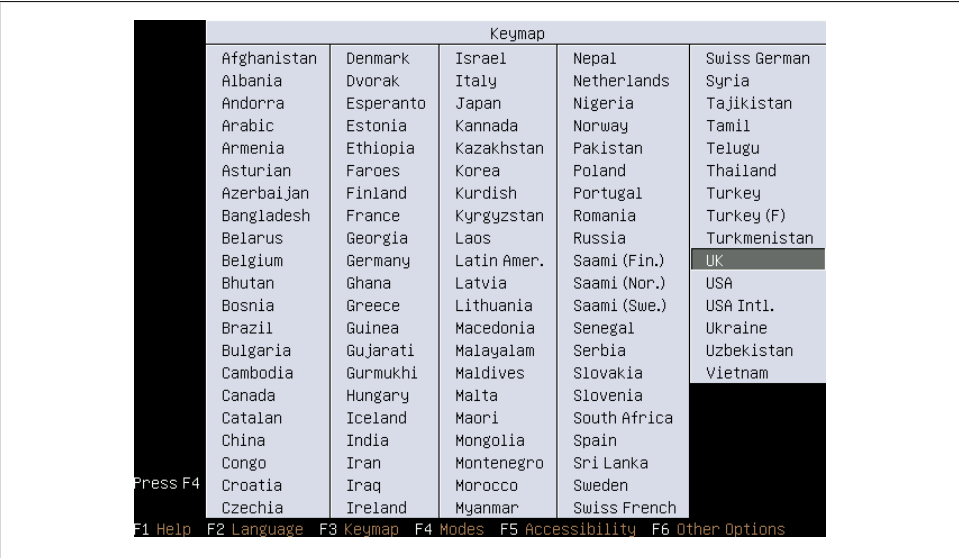


Figure 2-4. Choosing your country



Figure 2-5. Selecting the graphics mode to use

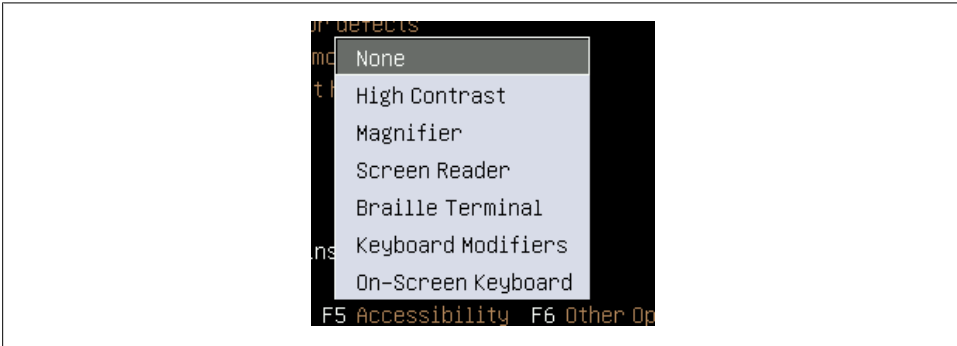


Figure 2-6. The accessibility features available

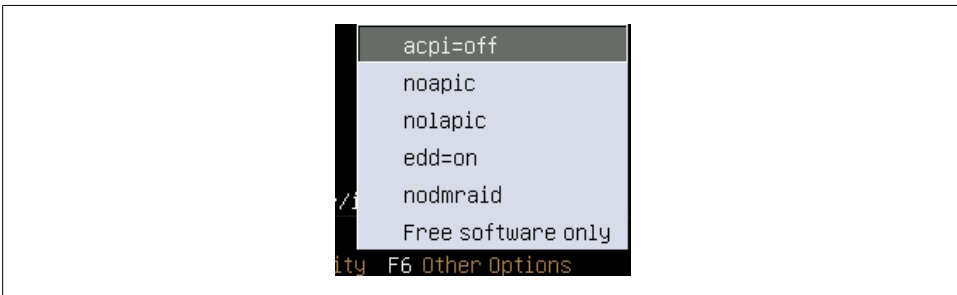


Figure 2-7. The Other Options menu

You can back out of any screen by pressing the Esc key.

Starting the Live DVD

Select the first choice from the menu, “Try Ubuntu without any change to your computer,” to start loading the operating system. When you do this, the screen will show a small Ubuntu logo in white, followed by a large, animated version.

In between you may see some text scroll by, but you can generally ignore this, even if you see what look like warning messages, as they usually are just generated when the

operating system checks your PC for certain types of nonessential hardware and does not find them.

After that, a small, animated disc will appear while Ubuntu loads the operating system drivers and other files into memory. Then, you'll see the screen shown in [Figure 2-8](#), which is the Ubuntu desktop.

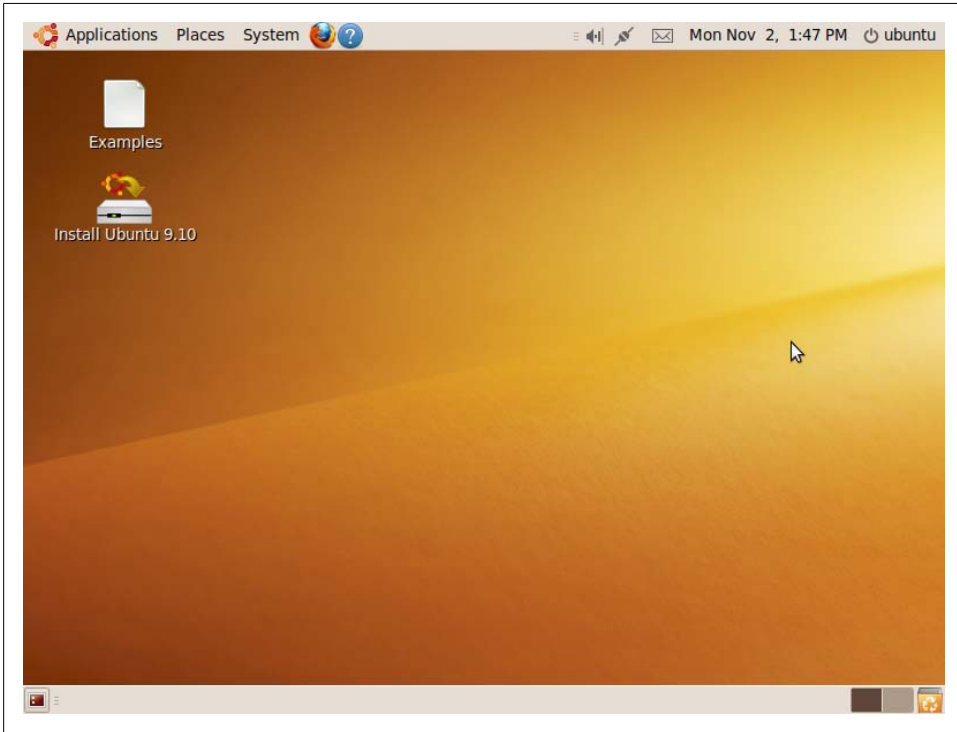


Figure 2-8. Ubuntu, up and running from the Live DVD

The various parts of the desktop are explained in the following chapters, but you may want to double-click the *Examples* icon to bring up a folder of examples and *readme* files that are worth browsing through (see [Figure 2-9](#)).

For example, the file *Welcome_to_Ubuntu.odt* is a document that you can double-click to view, as shown in [Figure 2-10](#). You can safely open all the files in this folder and its subfolders by double-clicking them. These include documents, spreadsheets, images, music, presentations, and more.

By opening a range of files from the *Examples* folder, you'll also be able to ensure that Ubuntu works properly on your computer, including whether your audio is functioning.

To determine whether your networking is operational, check for an icon in the top panel that looks like two plugs connecting together. If instead you see an icon that looks

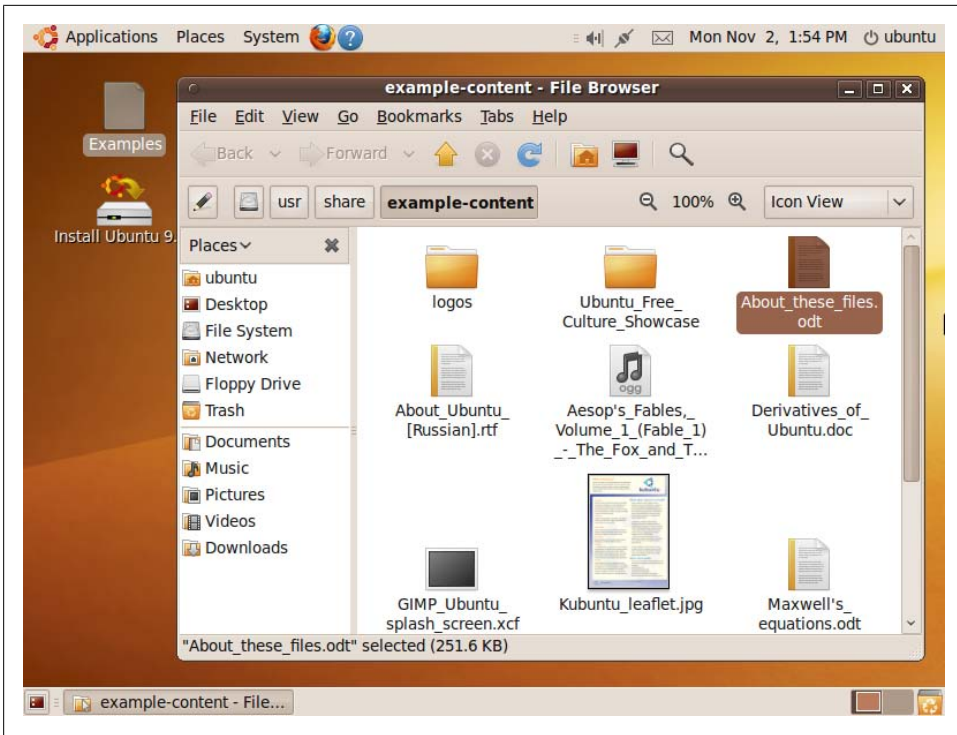


Figure 2-9. Browsing the Examples folder

like the “no signal” symbol on mobile phones (an antenna next to five dots), your networking is not currently working. Try right-clicking that icon and then ensure that the Enable Networking checkbox is checked. If it is and you have an Ethernet cable connected, the symbol showing two plugs should appear. If not, left-click the symbol and select “Auto eth0” to enable your networking card.

If you have a wireless networking card and it isn’t shown when you left-click that icon, you will need to establish a wired Ethernet connection in order to search for and locate a suitable driver. Details on how to do this are in [Chapter 3](#).



A long as you can connect via Ethernet and your mouse, keyboard, monitor, sound card, and other peripherals work using the Live DVD, installing Ubuntu shouldn’t present any problems. But as a newcomer to Ubuntu, you may want to install it alongside your computer’s original operating system until the time comes that you find you can do everything you need with it, and are prepared to delete your original OS to reclaim the space it uses.

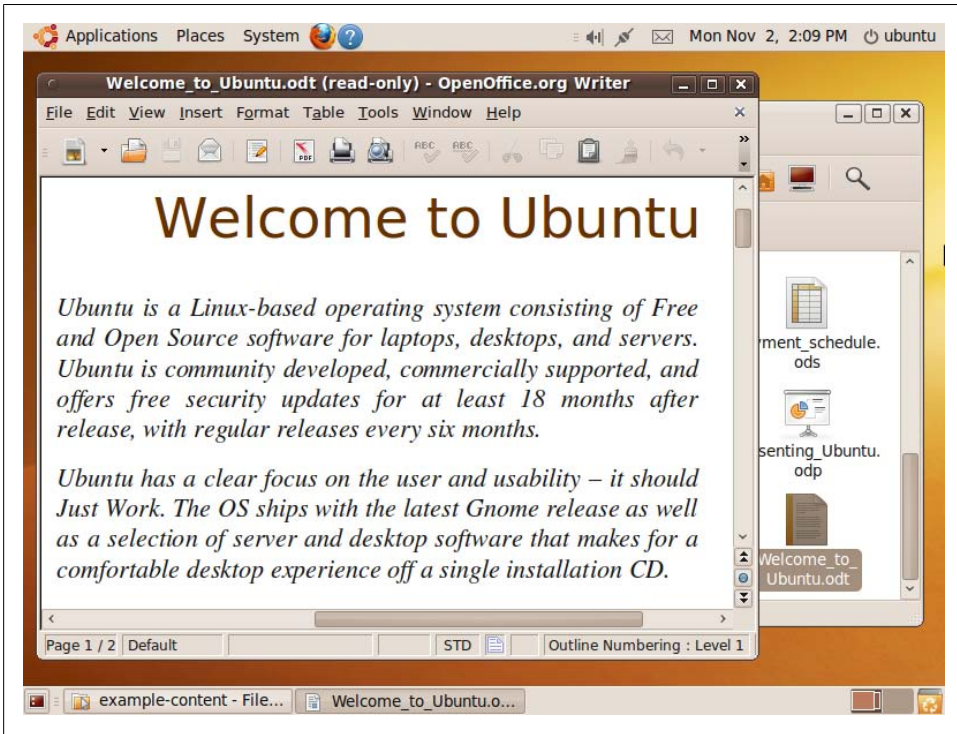
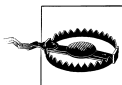


Figure 2-10. Viewing a document in OpenOffice.org

Alternative Editions

The DVD supplied with this book also contains a number of other Ubuntu editions that are saved as ISO files—representations of an entire CD-ROM that can be easily transferred across networks and the Internet. They will take up an entire CD when copied, and can also be opened and treated like optical discs if you have the right software.



Although the installation procedures for other varieties of Ubuntu are similar, they are not identical to the directions in this chapter for installing the standard desktop edition. In fact, there are so many distros that there's simply not enough room in this book to detail the installation process for each. Therefore, when installing an alternative edition, you should keep your eyes open, carefully read all the prompts and information provided during the process, and use your common sense when making choices and entering information.